Roll No. ....

TOE–801

B. TECH. (EE) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2023

EXPERT SYSTEM AND FUZZY LOGIC

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the following defuzzification methods used in fuzzy logic : (CO2, CO3)

(i) Center of largest area

(ii) First (or last) of maxima

(iii) Center of sums

(iv) Mean max. membership

(b) For the following navigation problem design a Mamdani fuzzy logic controller. Let us consider a problem scenario related to navigation of a mobile robot in presence of four moving obstacle. Obstacle 2 is found to be the most critical one. The aim is to develop a fuzzy logic-based motion planner that will be able to generate the collision free path for the robot. There are two inputs, namely the distance between the robot and the obstacle (D), and the angle $\left(\angle GSO_2\right)$ for the motion planner and it will generate one output, that is, deviation. Distance is represented using four linguistic terms, namely, very near (VN), near (NR), far (FR), and very far (VFR). Whereas the input (i.e., angle) and the output (i.e., deviation are expressed with the help of five linguistic terms such as Ahead (A). Ahead left (AL), Left (LT), Ahead right (AR) and Right (RT).

Determine the output. (CO2, CO3, CO4, CO5)

(c) For the problem scenario mentioned in question (b), determine the output if distance D = 1.04 m and angle $(\angle GSO_2) = 30$, using Mamdani approach. Use any *one* of the defuzzification method. (CO2, CO4, CO5)

2. (a) Explain different defuzzification methods used in fuzzy logic : (CO2, CO3)

(i) Max. membership principle

(ii) Centroid method

(iii) Weighted average method

(iv) Mean max. membership

(b) A rail road company intends to lay a new rail line in a particular part of a county. The whole area through which the new line is passing must be purchased for right-of-way considerations. It is surveyed in three stretches, and the data are collected for analysis. The surveyed data for the road are given by the sets, B1, B2, and *B3*. where the sets are defined on the universe of right-of-way widths, in metres. For the railroad to purchase the land, it must have an assessment of the amount of land to be bought. The three surveys on right-of-way width are ambiguous, however, because some of the land along the proposed railway route is already public domain and will not need to be purchased. Additionally, the original surveys are so old (*circa* 1860) that some ambiguity exists on boundaries and public right-of-way for old utility lines and old roads. The three fuzzy sets, B1, B2, and B3, shown in Figs. 1, 2 and 3, respectively, represent the uncertainty in each survey as to the membership of right-of-way width, in meters, in privately owned land.

We now want to aggregate these three survey results to find the single most nearly representative right-of-way width (z) to allow the railroad to make its initial estimate of the right-of-way purchasing cost. Use centroid method and weighted average method to find the single most nearly representative right-of-way width (z).

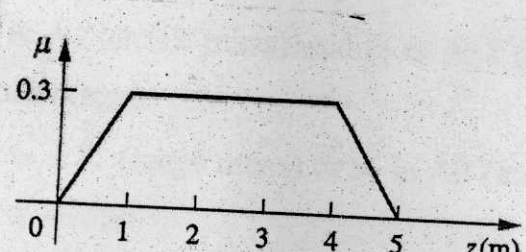

FIGURE-1

Fuzzy set B_1 : public right-of-way width (z) for survey 1.

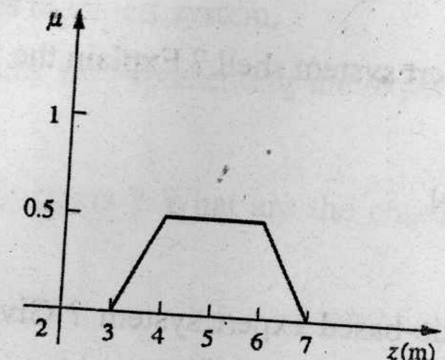

FIGURE-2

Fuzzy set B_2 : public right-of-way width (z) for survey 2.

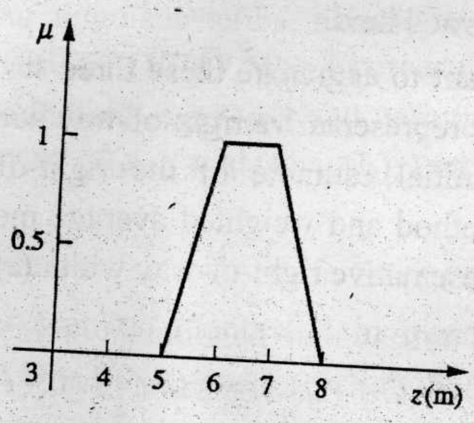

FIGURE-3

Fuzzy set B_3 : public right-of-way width (z) for survey 3.

(c) Draw the block diagram of fuzzy logic controller. Explain the function of each component. Also give an suitable example. (CO2)

3. (a) Draw the general structure of an expert system. Also explain its components. (CO1)

(b) What is a expert system shell ? Explain the following : (CO1)

(i) MYCIN

(ii) EMYCIN

(iii) COL

(c) What is a rule-based expert system ? Give *one* example of rule based reasoning in expert system. (CO1, CO5)

4. (a) Why should we use Expert Systems ? What are the limitations of expert systems ? (CO1)

(b) In mechanics, the energy of a moving body is called kinetic energy. If an object of mass m (kilograms) is moving with a velocity v (metres per second), then the kinetic energy k (in joules) is given by the equation $k = \frac{1}{2}mv^2$. Suppose we model the mass and velocity as inputs to a system (moving body) and the energy as output, then observe the system for a while and deduce the following two disjunctive rules of inference based on our observations : (CO4)

Rule 1: IF x_1 is A1 (small mass) and x_2 is A12 (high velocity), THEN y is B1 (medium energy).

Rule 2 : IF x_1 is A21 (large mass) or x_2 is A2 (medium velocity), THEN y is B2 (high velocity).

Fire the above two rule to give one fuzzy output. Assume the membership function for A1, A21, A12, A2, B1, and B2.

(c) What is database and rule base in fuzzy logic system ? Explain the fuzzy reasoning process with an suitable example. (CO4)

5. (a) What is knowledge elicitation ? Explain the different knowledge elicitation techniques in expert system. (CO1)

(b) Explain the techniques for representing the expert knowledge in expert system. (CO2)

(c) What is an Expert Systems ? What are the characteristics/attributes of Expert systems ? (CO2)

Roll No. ....

TEE-802

B. TECH. (EE) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2022

WIND AND SOLAR ENERGY SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the drivers for changes in renewable energy system ? Discuss in detail. (CO2, CO6)

   (b) Discuss the role of smart grid for sustainable development with neat diagram. (CO2, CO6)

   (c) Compare between smart and conventional grid. (CO2, CO6)

2. (a) Discuss flat plate type solar collectors and concentrating focusing type solar collector. (CO5, CO4)

   (b) What are the main issues of existing power system ? Discuss its remedies in term of smart grid. (CO5, CO4)

(c) Discuss the smart controllers used in automation of smart grid.

(CO5, CO4)

3. (a) What is a photovoltaic cell ? Explain briefly with neat diagram of a silicon PV cell. (CO1, CO2)

(b) What is solar pond ? Discuss its principle and working with neat diagram. (CO1, CO2)

(c) State and explain the differences between active and passive space heating system. (CO1, CO2)

4. (a) Discuss the advantages and disadvantages of wind energy conversion system. (CO1, CO3)

(b) Draw solar PV characteristics and discuss its various salient points.

(CO1, CO3)

(c) Draw the block diagram of principal component of standalone PV system and explain. (CO1, CO3)

5. (a) What are the advantages and limitations of solar system ? Discuss its applications. (CO5, CO6)

(b) What is the basis of wind energy conversion ? Explain. (CO5, CO6)

(c) What are the basic component of wind energy conversion system ? Explain them with neat block diagram. (CO5, CO6)

50

Roll No. ....

TOE–802

B. TECH. (EE) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2023

INTELLIGENT SENSORS AND INSTRUMENTATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Why semiconductors are used as sensors? What do you mean by semiconductor sensor ? What it does ? Mention the advantages of semiconductor sensor ? (CO1, CO2)

   (b) Explain the working principle of MEMS sensors. Mention the examples of MEMS sensors. Discuss the advantages of MEMS sensors. Where are they used ? Give examples. (CO1, CO2)

   (c) Discriminate the actuators from sensors. Discuss the working principle of actuator. Give the few examples of actuators. (CO1, CO2)

2. (a) What is a smart sensor ? How does it differ from sensors ? What are the types of smart sensors ? List out the few applications of smart sensors. (CO1, CO2)

   (b) What do you understand by neurosensors ? Briefly explain neuromorphic vision sensor.

   (c) Write a brief note on biosensors. (CO1, CO2)

3. (a) Describe thin and thick film deposition techniques for the fabrication of sensors. (CO3, CO4)

   (b) Describe the wet and dry etching technique of sensors material. (CO3, CO4)

   (c) Write a brief note on sensor material. (CO3, CO4)

4. (a) What is instrumentation ? What exactly it does ? How does the intelligent instrumentation differ from instrumentation ? (CO3, CO4)

   (b) What is Fuzzy ? What is fuzzy rule base ? What do you mean by crisp value in fuzzy ? Briefly explain fuzzy inference engine and defuzzification. (CO3, CO4)

   (c) Write brief notes on GA, ANN and Machine Learning. (CO3, CO4)

5. (a) What is IEEE 1451 standard? What is IEEE 1451.4 standard? Expand TEDS and STIM. (CO5)

   (b) How does the Machine Learning differ from Genetic Algorithm ? Explain clearly. (CO5)

   (c) What is NAC? What are the key components of NAC ? What is the main steps/process while implementing NAC ? (CO5)

2. (a) What is a smart sensor ? How does it differ from sensors ? What are the types of smart sensors ? List out the few applications of smart sensors. (CO1, CO2)

   (b) What do you understand by neurosensors ? Briefly explain neuromorphic vision sensor.

   (c) Write a brief note on biosensors. (CO1, CO2)

3. (a) Describe thin and thick film deposition techniques for the fabrication of sensors. (CO3, CO4)

   (b) Describe the wet and dry etching technique of sensors material. (CO3, CO4)

   (c) Write a brief note on sensor material. (CO3, CO4)

4. (a) What is instrumentation ? What exactly it does ? How does the intelligent instrumentation differ from instrumentation ? (CO3, CO4)

   (b) What is Fuzzy ? What is fuzzy rule base ? What do you mean by crisp value in fuzzy ? Briefly explain fuzzy inference engine and defuzzification. (CO3, CO4)

   (c) Write brief notes on GA, ANN and Machine Learning. (CO3, CO4)

5. (a) What is IEEE 1451 standard? What is IEEE 1451.4 standard? Expand TEDS and STIM. (CO5)

   (b) How does the Machine Learning differ from Genetic Algorithm ? Explain clearly. (CO5)

   (c) What is NAC? What are the key components of NAC ? What is the main steps/process while implementing NAC ? (CO5)